# Regulatory & Risk Framework Primer for Quants

The Hidden Geometry of Capital and Model Design

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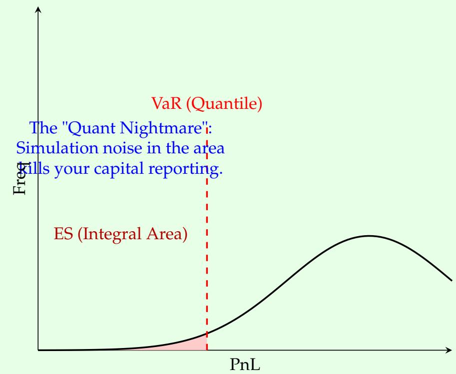


Figure 4: The visual difference between 'finding a point' (VaR) and 'integrating a tail' (ES).

**The 3-Second Answer (Module 3)**

**Question:** *"Why move from 99% VaR to 97.5% ES?"*
**Your opening line:**

Because VaR is **blind to tail depth** and violates **sub-additivity** (diversification benefit). 97.5% ES ensures banks hold capital for the 'average' loss in the tail, forcing a bigger buffer for fat-tailed exotic positions.

### 3.3 FRTB: The "Capital Toxic" Boundaries

FRTB (Fundamental Review of the Trading Book) introduced structural changes:

- **Liquidity Horizons:** Instead of a flat 10-day VaR, different assets have horizons from 10 to 120 days. (FX = 10d, Small-Cap Equity = 60d).
- **The Boundary:** Much harder to move assets between the **Banking Book** (Accrual) and **Trading Book** (Mark-to-Market).

### 3.4 Case: Structured Note — VaR vs ES

Imagine a "Digital Option" that pays out $100M if a crash exceeds 5%, but $0 otherwise.

- **Under VaR (99%):** If the crash happens at the 99.5% line, the 99% VaR might be $0. (Invisible risk).
- **Under ES (97.5%):** The ES will average all losses above 97.5%, capturing a large portion of that $100M payout.

**Result:** Products with "cliff risk" became **Capital Toxic** overnight.

**Practitioner Insight**

Desk Strategy: Avoid digital-style payoffs. Smooth out your tails to lower your ES capital charge.

**Interview Question**

**Question:** Why did the industry move from 99% VaR to 97.5% ES?
**Context:** Testing understanding of coherent risk measures.
**Answer:** 97.5% ES provides a similar level of capital to 99% VaR for "normal" distributions, but much higher capital for "fat-tailed" ones. ES is a **coherent risk measure** (sub-additive), whereas VaR is not. This move forces banks to explicitly reserve for "Black Swan" events rather than just the first point of the tail.

18. **Define 'Specific Risk'.** *Ans:* The risk unique to a single issuer (e.g., corporate default) that can be diversified away, unlike "General Market Risk".
19. **Why use 'Importance Sampling'?** *Ans:* To force the Monte Carlo simulator to sample "rare" tail events more frequently, reducing the variance of the ES estimator.
20. **Explain 'Basis Risk' in SOFR.** *Ans:* The risk arising from the mismatch between the new SOFR rate and legacy LIBOR-linked instruments or different SOFR tenors.
21. **What is 'NSFR'?** *Ans:* Net Stable Funding Ratio. It ensures banks fund long-term assets with reliable, long-term liabilities to prevent liquidity crunches.
22. **What is 'Model Inventory'?** *Ans:* A centralized database of all models, their risk ratings, validation status, and known limitations.
23. **Explain 'SMA' for Operational Risk.** *Ans:* The Standardized Measurement Approach, which replaces the complex internal AMA for OpRisk with a logic based on business size and internal loss history.
24. **Define 'Jump-Diffusion'.** *Ans:* A model (like Merton's) that combines continuous Brownian motion with sudden "jumps" to capture market crashes or single-name defaults.
25. **What is the 'IMM' approval?** *Ans:* Internal Model Method. It is the holy grail for a bank—the regulator's permission to use their own exposure models for Counterparty Risk capital.

### 13.2 Pressure Questions: The "Quant Stress Test"

These questions test your ability to handle operational, ethical, and high-stakes desk dilemmas.

1. **"You find a fatal flaw in the bank's core RWA engine on the last day of reporting. The CEO has already signed the filing. What is your immediate next step?"** *Ans:* Immediately notify the Head of Model Risk and the reporting team. Transparency is the only defense. A "Late Filing" or an "Amended Filing" is painful, but a "False Filing" is a criminal regulatory violation.
2. **"The Desk Head wants to trade an exotic product that the Validator says is 'Unmodellable'. High profit, high capital. Who do you side with?"** *Ans:* You side with the **Validator**. While profit is good, an unmodellable product carries "Hidden Tail Risk" that the bank cannot price accurately. If you side with the desk, you are compromising the bank's solvency for a single bonus cycle.
3. **"A competitor's model predicts 20% less capital for the same book. Are they better quants or are they cheating? How do you investigate?"** *Ans:* Internal benchmarking. I would attempt to replicate their results using their public disclosures. Often, the difference isn't 'better math' but different **Modellability Assumptions** or a more aggressive **Netting Logic**. I would present the findings to the desk as a "Strategy Gap" analysis.
4. **"Your backtesting engine shows 12 exceptions (Red Zone). You suspect the data feed was wrong, not the model. How do you prove this to the Regulator?"** *Ans:* I perform a **Clean vs Hypo P&L** analysis. If the exceptions only occur in Real P&L (due to data noise/fees) but not in Hypo P&L (model only), then the model is sound. I would document the specific data anomalies (e.g., stale prices) as the root cause.
5. **"If you had to explain 'Stochastic Calculus' to a Compliance Officer with zero math background to justify a $100M capital release, what analogy do you use?"** *Ans:* Use the **"Random Walk on a Tightrope"** analogy. Stochastic Calculus is the math that tells us how far the walker can sway before falling. By using a better model, we are proving that our "safety net" (capital) is currently 10x larger than the worst possible sway, allowing us to safely reduce it while staying secure.

## Appendix L: Python Snippet — The 'Quant Mini-Library'

To pass a technical interview, you should be able to sketch the logic of a regulatory engine.

**Practitioner Insight**

**The ES Calculation (Pythonic Logic):**

```
import numpy as np

def calculate_es(pnl_paths, alpha=0.975):
    # Sort paths from worst to best
    sorted_pnl = np.sort(pnl_paths)

    # Identify the VaR index
    var_index = int((1 - alpha) * len(sorted_pnl))

    # VaR is the threshold
    var = sorted_pnl[var_index]

    # ES is the mean of all losses worse than VaR
    es = np.mean(sorted_pnl[:var_index])

    return var, es
```

**Note:** In production, you would use **GPU-acceleration (CUDA)** and **Importance Sampling** to handle the tail integration efficiently.

### L.1 SA-CCR Add-on Logic

```
def fx_addon(notional, maturity, sf=0.04):
    # Maturity Factor (MF)
    mf = np.sqrt(min(maturity, 1.0) / 1.0)

    # Effective Notional
    eff_notional = notional * sf * mf

    return eff_notional
```

This simple logic is the heart of every bank's "Standardized" regulatory reporting engine.